Course - EEE 4418

AUTOCAD · ELECTRICAL DESIGN PROJECT

ELECTRICAL LAYOUT DESIGN OF A FOUR STORIED RESIDENTIAL BUILDING



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INTRODUCTION

Project at a Glance

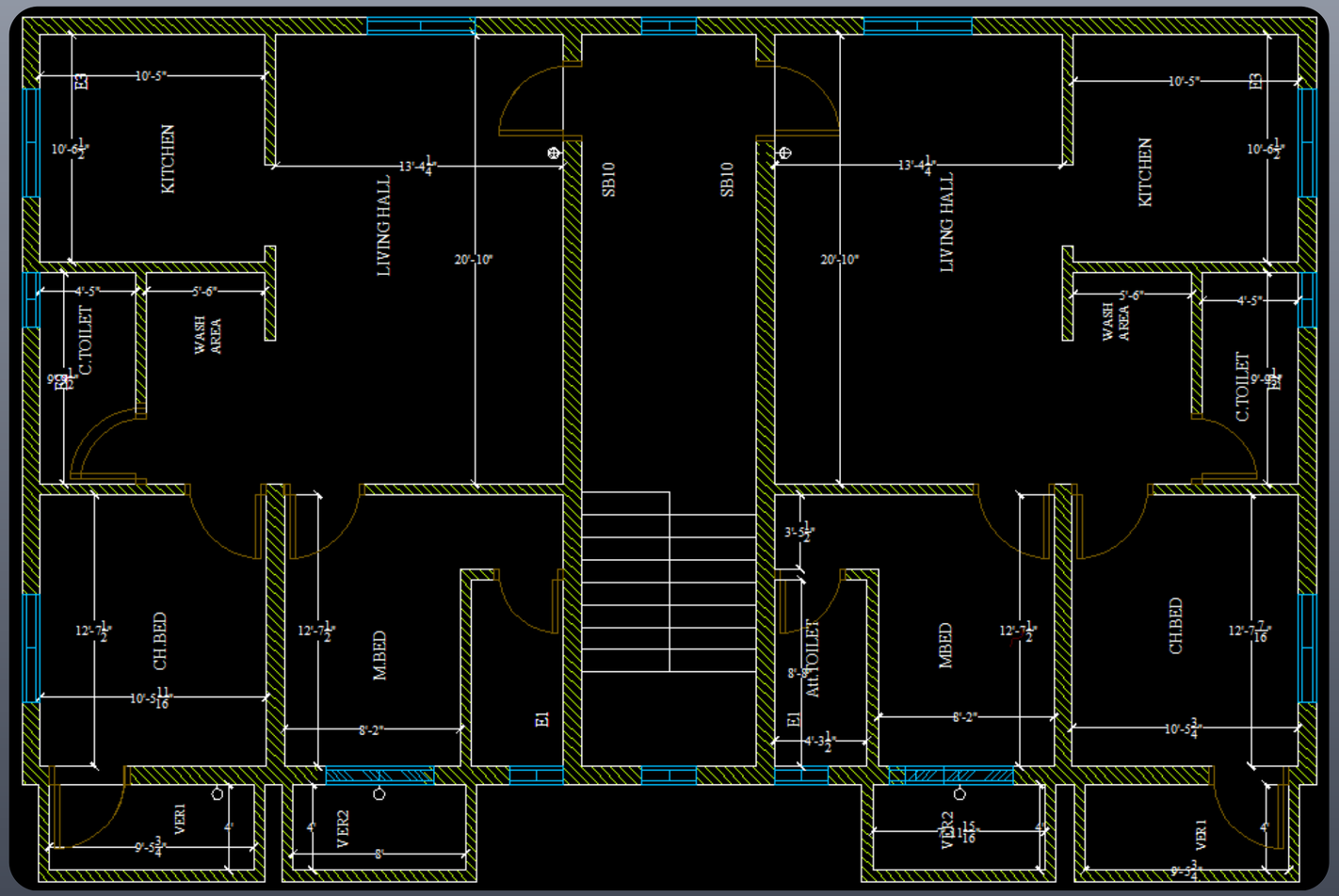
04	02	08	21	61.6KW
FLOORS	UNITS / FLOOR	TOTAL UNITS	BRANCH BOARDS	TOTAL LOAD

BUILDING & ROOM DATA

Building	60' × 35'-7.5"
footprint	33'-11.5"×8'-1.5"
Central lobby	12'-7.5" × 8'-2"
Master bed	12'-7.5" × 10'- 5. 7"
Child bed	20'-10" × 13'- 4.25"
Living hall	10'-6.5" × 10'-5" 8'-8"
Kitchen	× 4'-3.5"
Attached toilet	9'-9.5" × 4'-5"
Common toilet	

ARCHITECTURAL PLAN

Typical Floor Plan



Typical floor

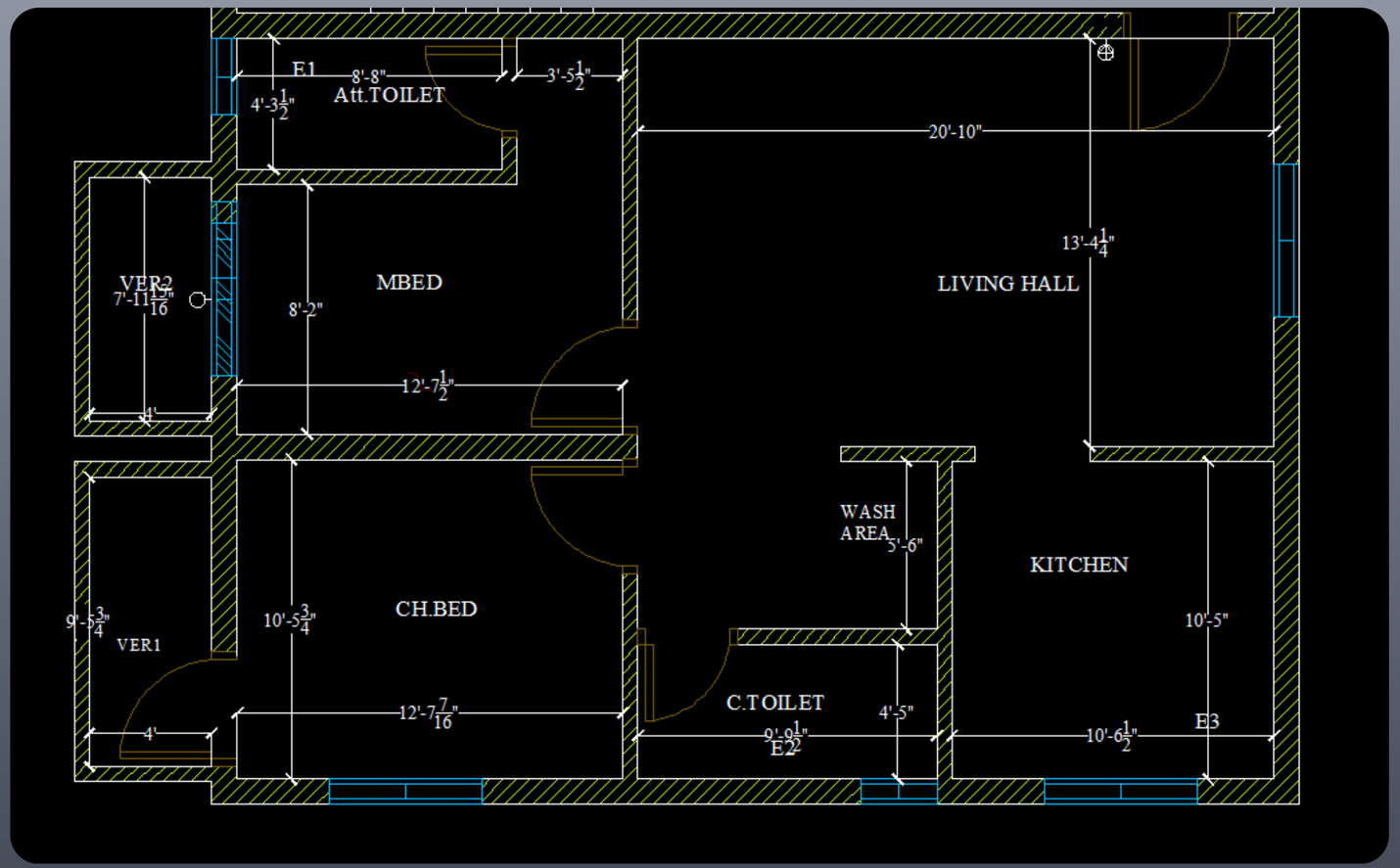


KEY FEATURES

Units per floor	2
Lobby	33'-11.5"×8'-1.5"
Overall	60' × 35'-7.5"

ARCHITECTURAL PLAN

Single Unit — Unit A



Representative of all two units on every floor

ROOM SCHEDULE

Master bed	12'-7.5" × 8'-2"
Child bed	12'-7.5" × 10'- 5. 7"
Living hall	20'-10" × 13'- 4.25"
Kitchen	10'-6.5" × 10'-5"
Attached toilet	8'-8" × 4'-3.5"
Common toilet	9'-9.5" × 4'-5"

CIRCULATION NOTES

Verandah attached to the master bedroom and Child bedroom

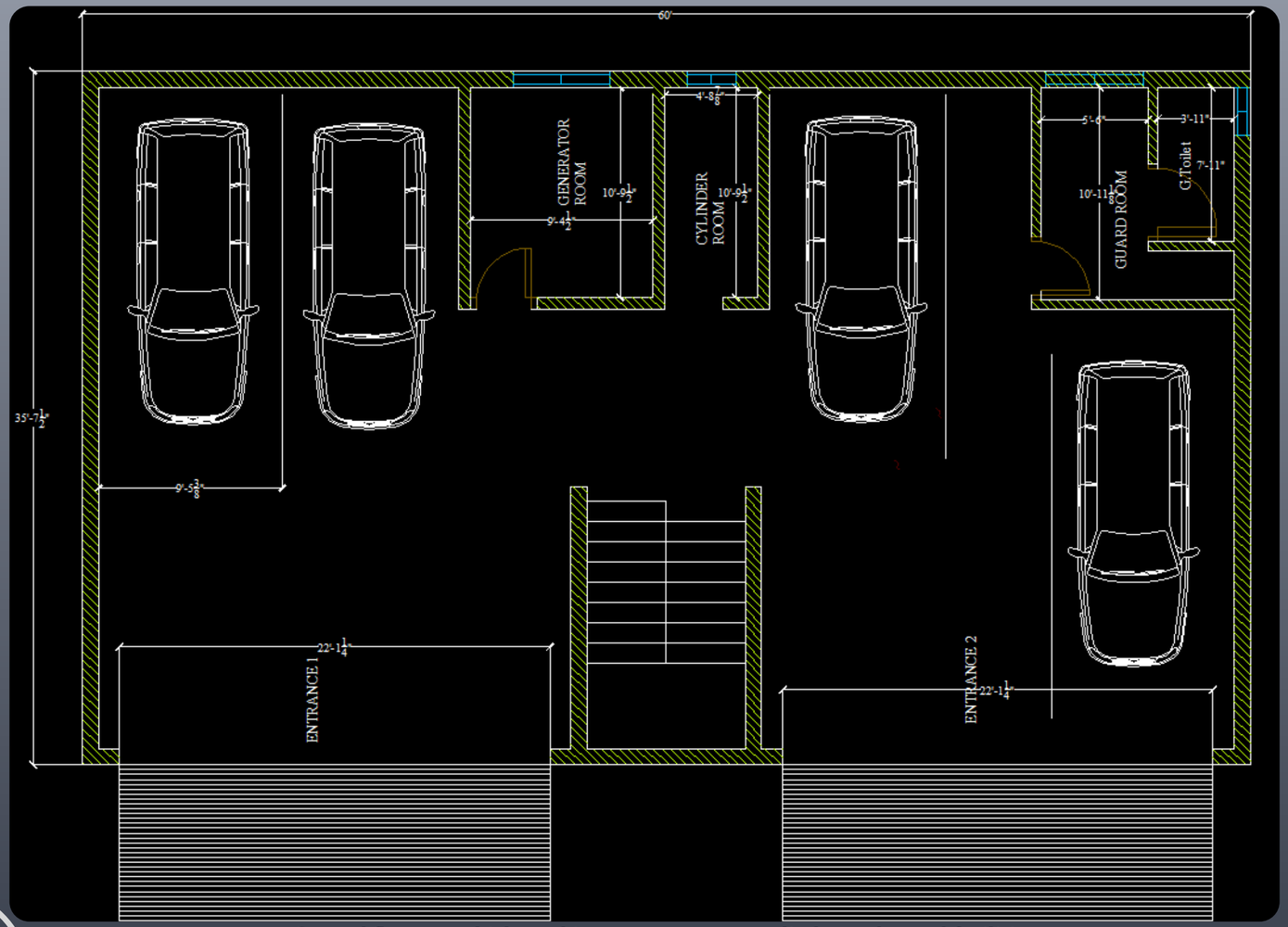
Kitchen placed on the outer wall for cross ventilation and exhaust

Attached toilet serves the master bed; common toilet serves the child bed

Living hall acts as the distribution space of the unit

ARCHITECTURAL PLAN

Ground Floor — Parking & Services



Ground floor — vehicle parking, generator, gas cylinder and guard facilities

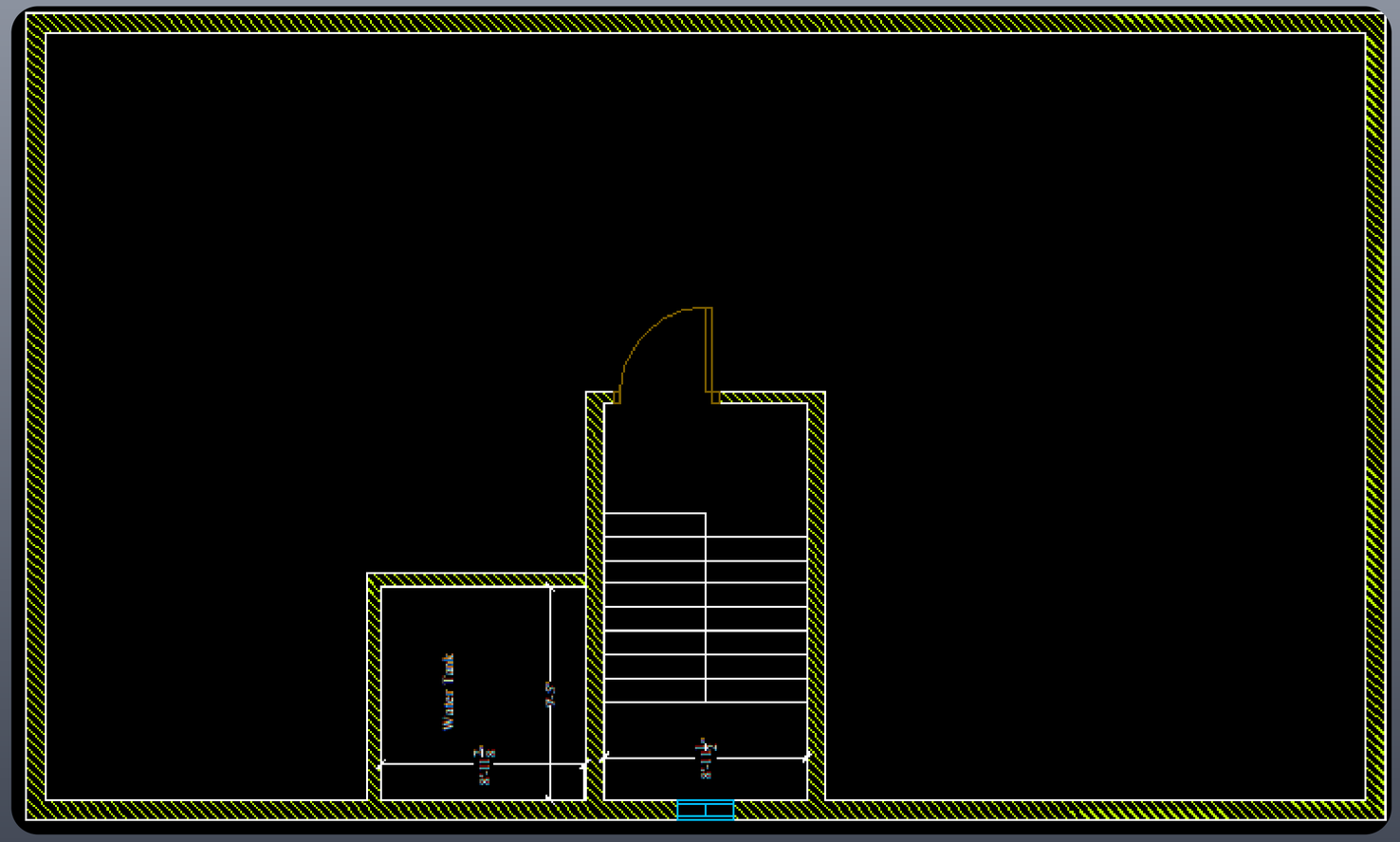
GROUND FLOOR PROVISIONS

Car parks	**16' × 9'-5.4"**
Generator room	**10'-9.5" × 9'-4.5"**
Gas cylinder room	**10'-9.5" × 4'-8.9"**
Guard room, toilet	**10'-11.1" × 5'-6", 7'-11" × 3'-11"**

- Stairs are situated on the west sides of the bulding.
- Ramp on the west side gives vehicle access to the parking level
- Generator and cylinder rooms are placed far away from the guard room for accommodation purposes.
- Guard room has a side window and the toilet has an exhaust window ensuring proper ventilation

ARCHITECTURAL PLAN

Roof Top Plan



ROOF TOP PROVISIONS

Roof top free area	**60' × 35'-7.5" -water tank and stair area**
Water tank Area	**9'-5"× 8'-11.9"**

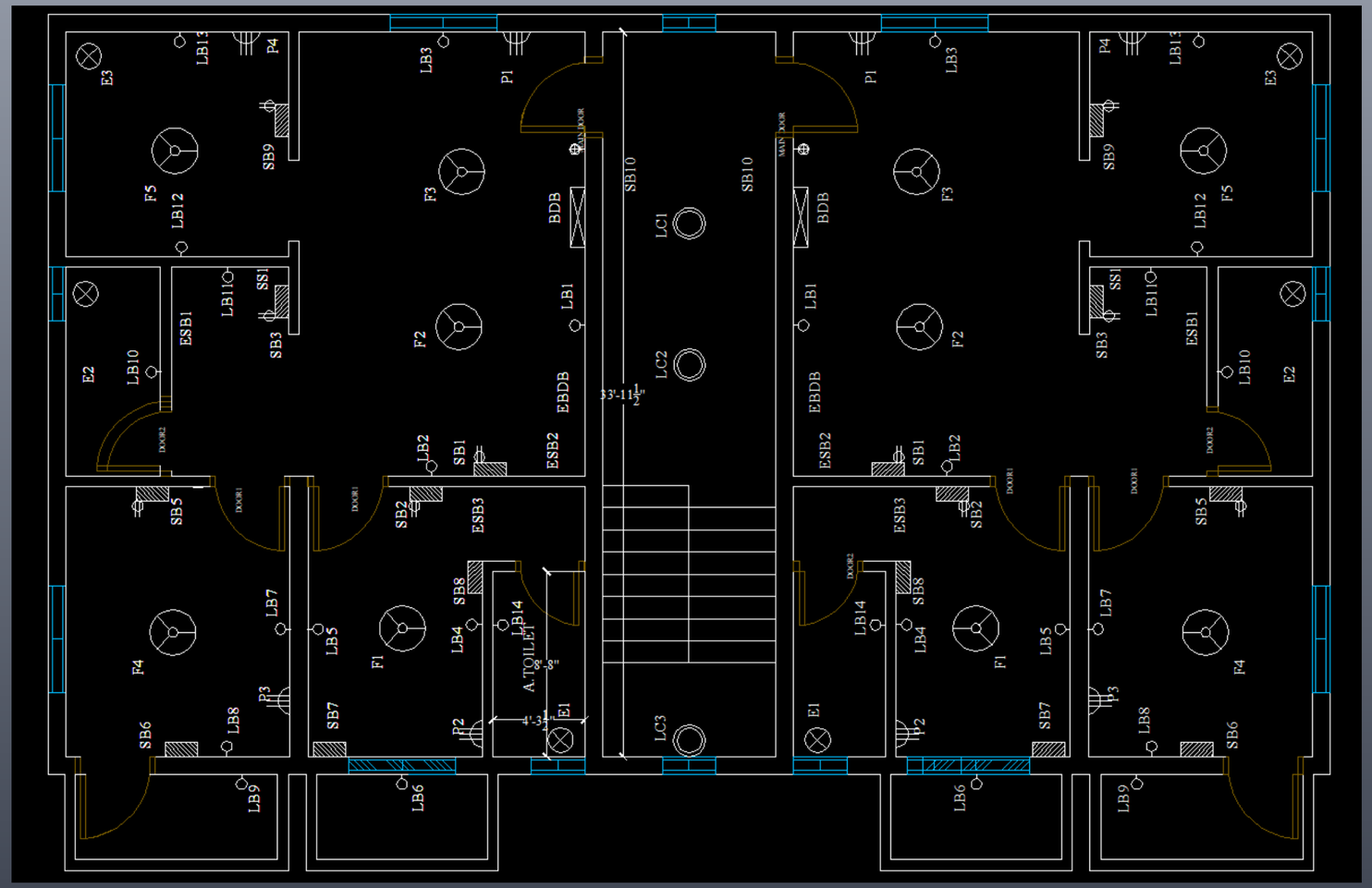
DESIGN NOTES

- Stair terminates in a covered stair head giving access to the open roof
- Remaining roof area left open for common use



ELECTRICAL LAYOUT

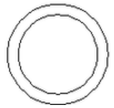
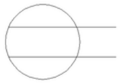
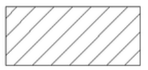
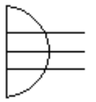
Fittings & Fixtures — Typical Floor



Typical floor — position of all lights, fans, switch boards and socket outlets

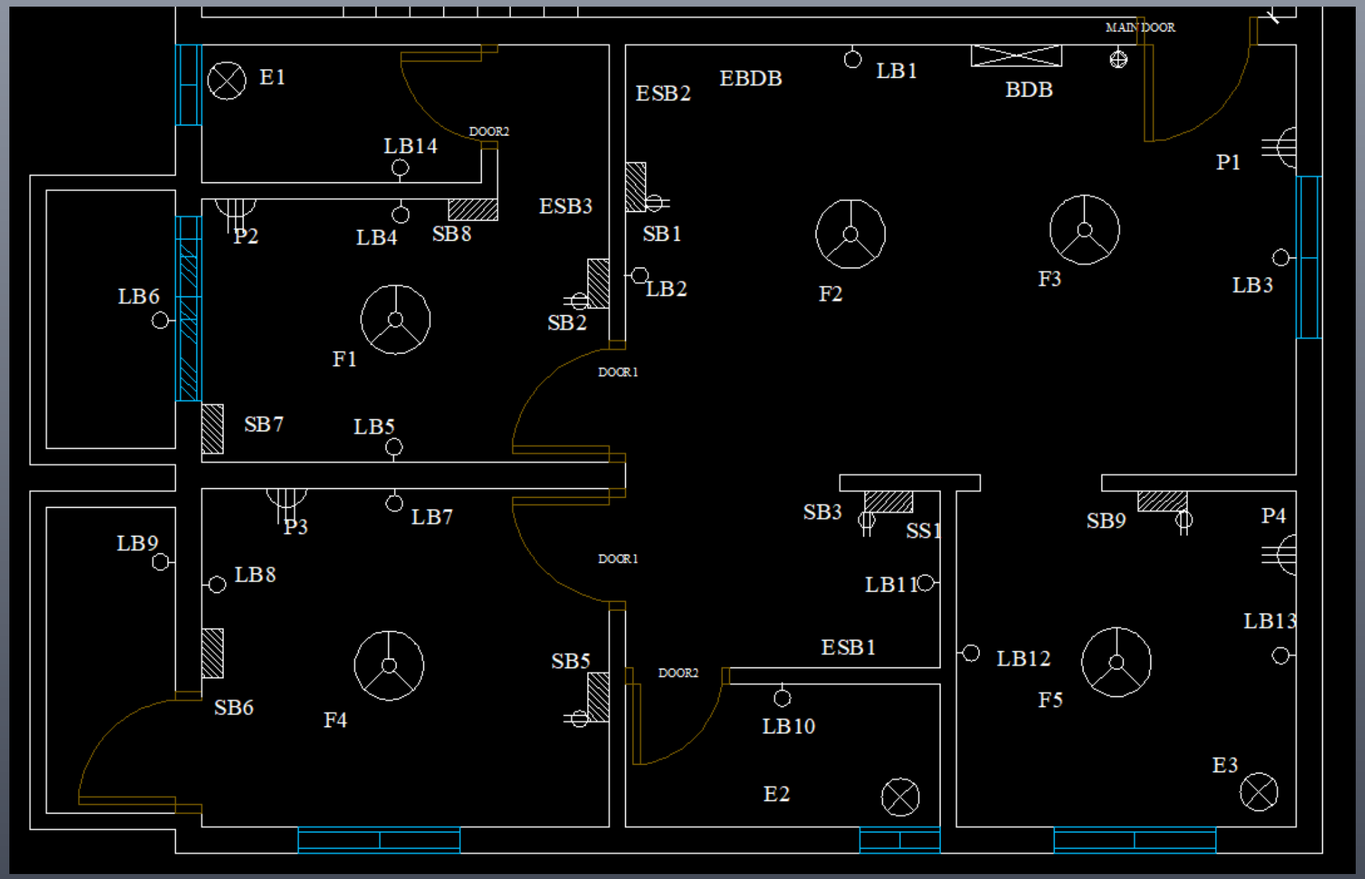


SYMBOL LEGEND

F		**Ceiling fan**
LB		**Light bulb**
CB		**Ceiling bulb**
SB		**Switch board**
SS		**Switchboard socket 2-pin socket outlet**
E		**Exhaust fan**
ESB		**Emergency board**
P socket		**P (3000w 15A) 3 pin sockets**

ELECTRICAL LAYOUT

Fittings & Fixtures — Single Unit



Unit A — enlarged view of the fitting and fixture layout

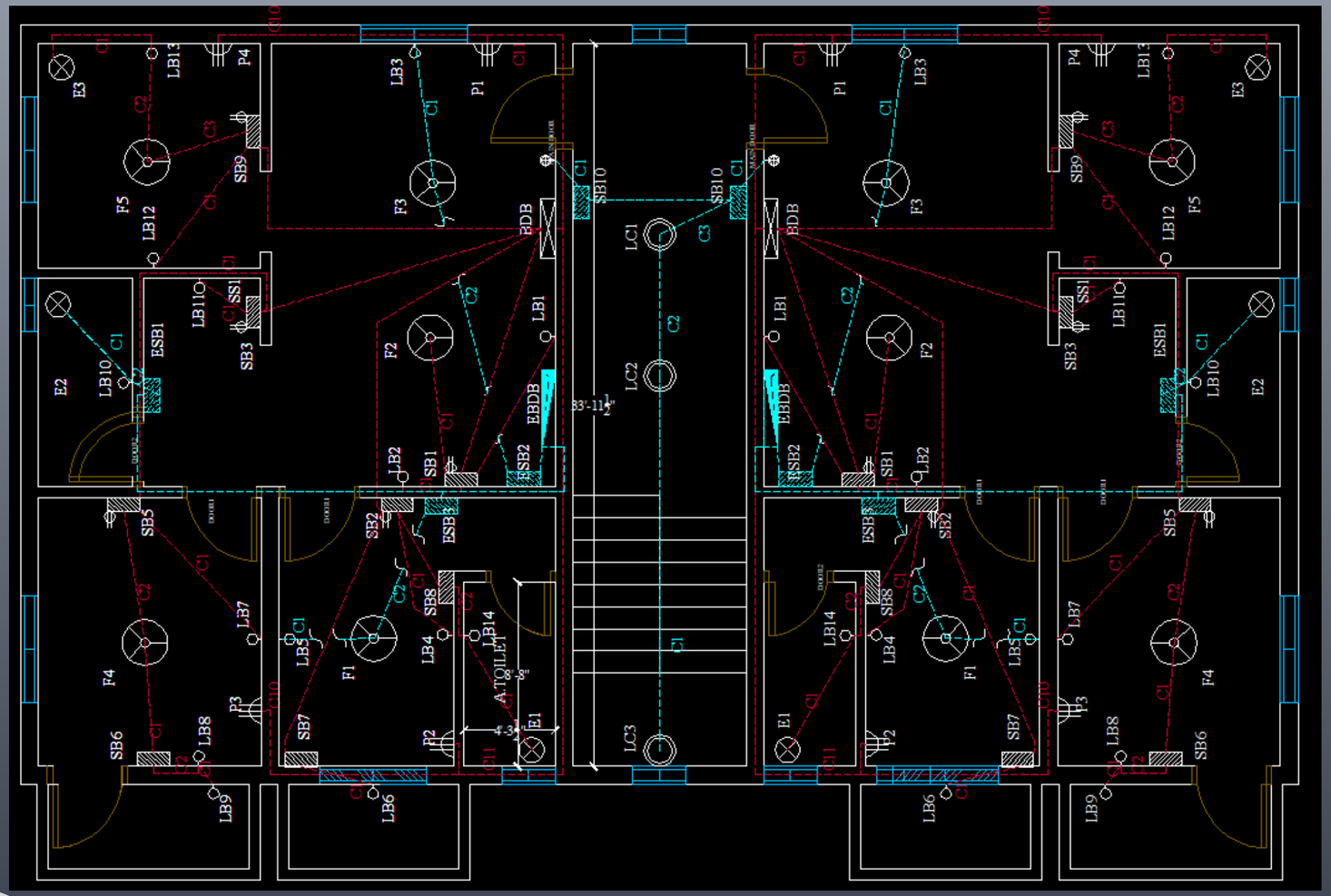
Room	Light	Fan	Switch board	3-pin socket	TV	Exhaust Fan
Master bed	2	1	1	1	0	0
Child bed	2	1	1	1	0	0
Living Hall	3	2	1	1	1	0
Kitchen	2	1	1	1	0	1
Att. Toilet	1	0	1	0	0	1
C. Toilet	1	0	1	0	0	1
Total Veranda	2	0	2	0	0	0
Total per unit	**11**	5	8	4	**1**	3

BOARD ARRANGEMENT

- Each room,veranda and toilet has its seperate switch boards in order to control the respective room's applinaces
- Every switch board carries one 2-pin socket outlet
- Every room contains a 3 pin 15A socket outlet for heavier appliances

WIRING DESIGN

Wiring Layout — Typical Floor



Typical floor — complete conduit and circuit routing for all four units

WIRING NOTES

Red lines	**Normal supply**
Cyan lines	**Emergency**
System	**220 V, 1-φ**

Conduit Specifications

C1==2x1.5 rm BYM

C2=4x1.5 rm BYM

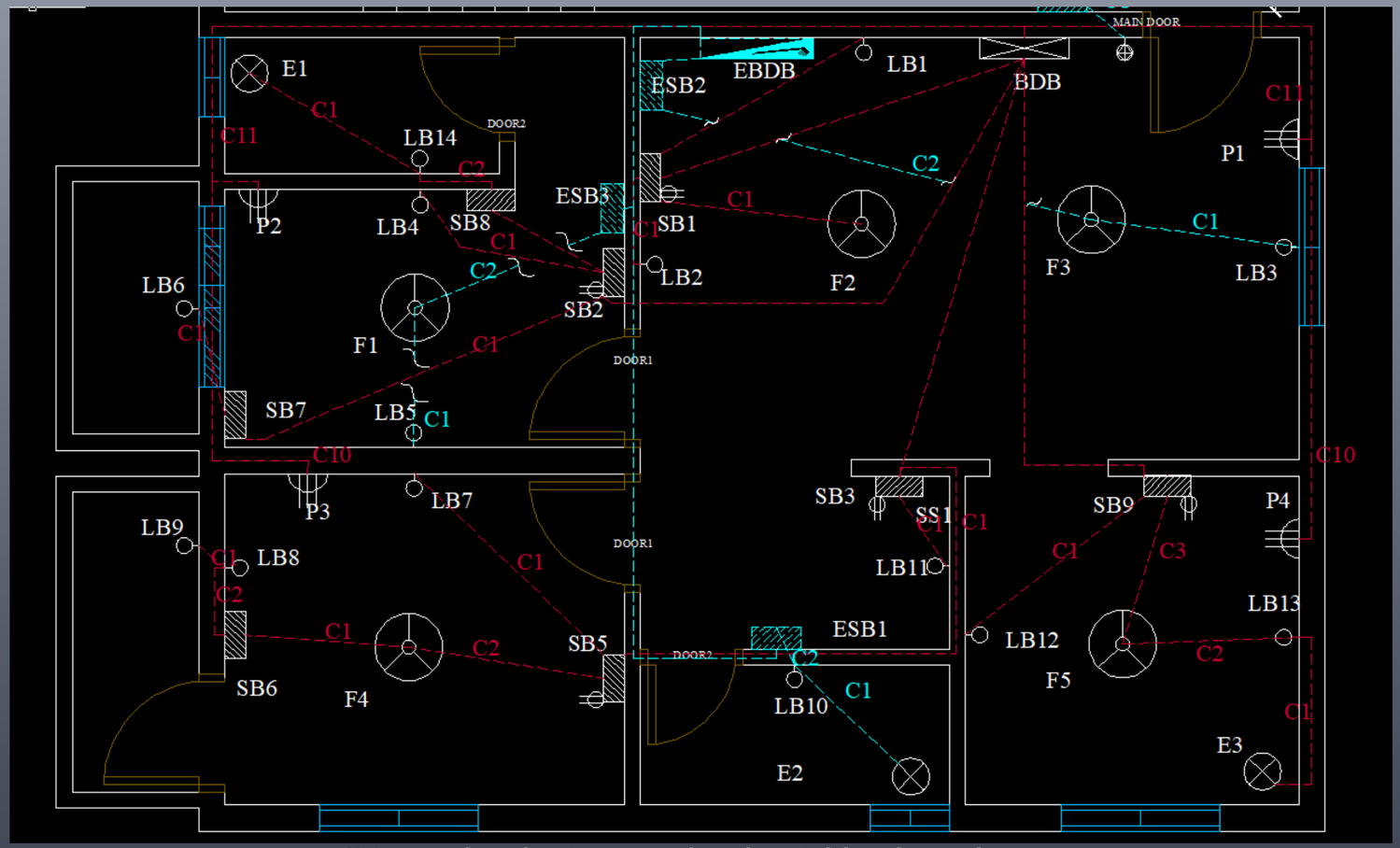
C3=6x1.5 rm BYM

C10= 2 x 6rm BYM + 6rm BYA ECC

C11= 4 x 6rm BYM + 6rm BYA ECC

WIRING DESIGN

Wiring Layout — Single Unit



Unit A — enlarged wiring routing from the switch boards to each point

CIRCUITS IN ONE UNIT

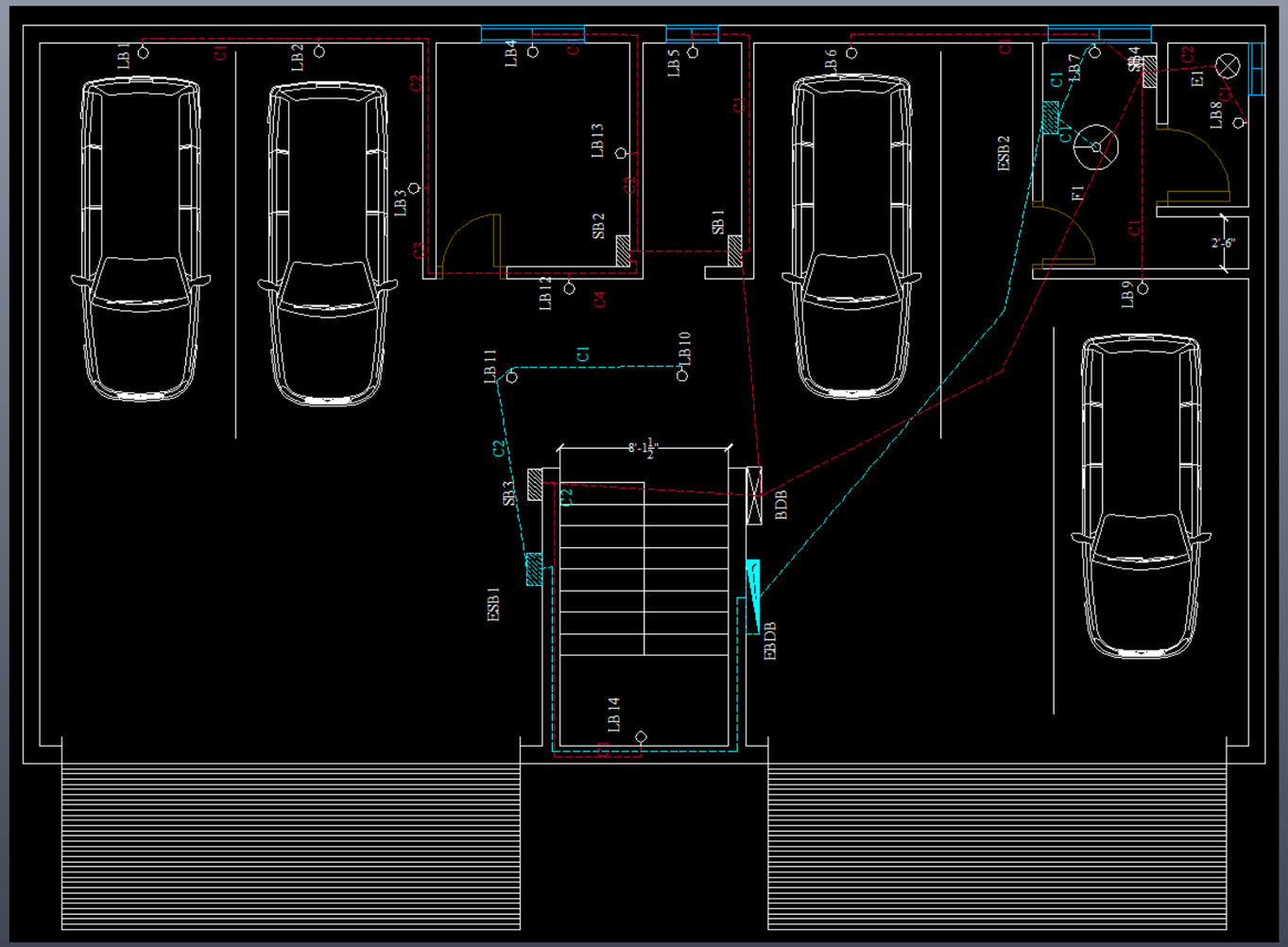
Normal circuits	**CKT-1 to CKT-4**
Socket circuits	**P1 to P4**
Switch boards	**SB1 – SB9**
Emergency boards	**ESB1 – ESB3**

ROUTING PRINCIPLE

One circuit serves one room group so a fault isolates only that area

Loop-in system used at the switch board — no joints inside the conduit

Wiring Layout — Parking Floor



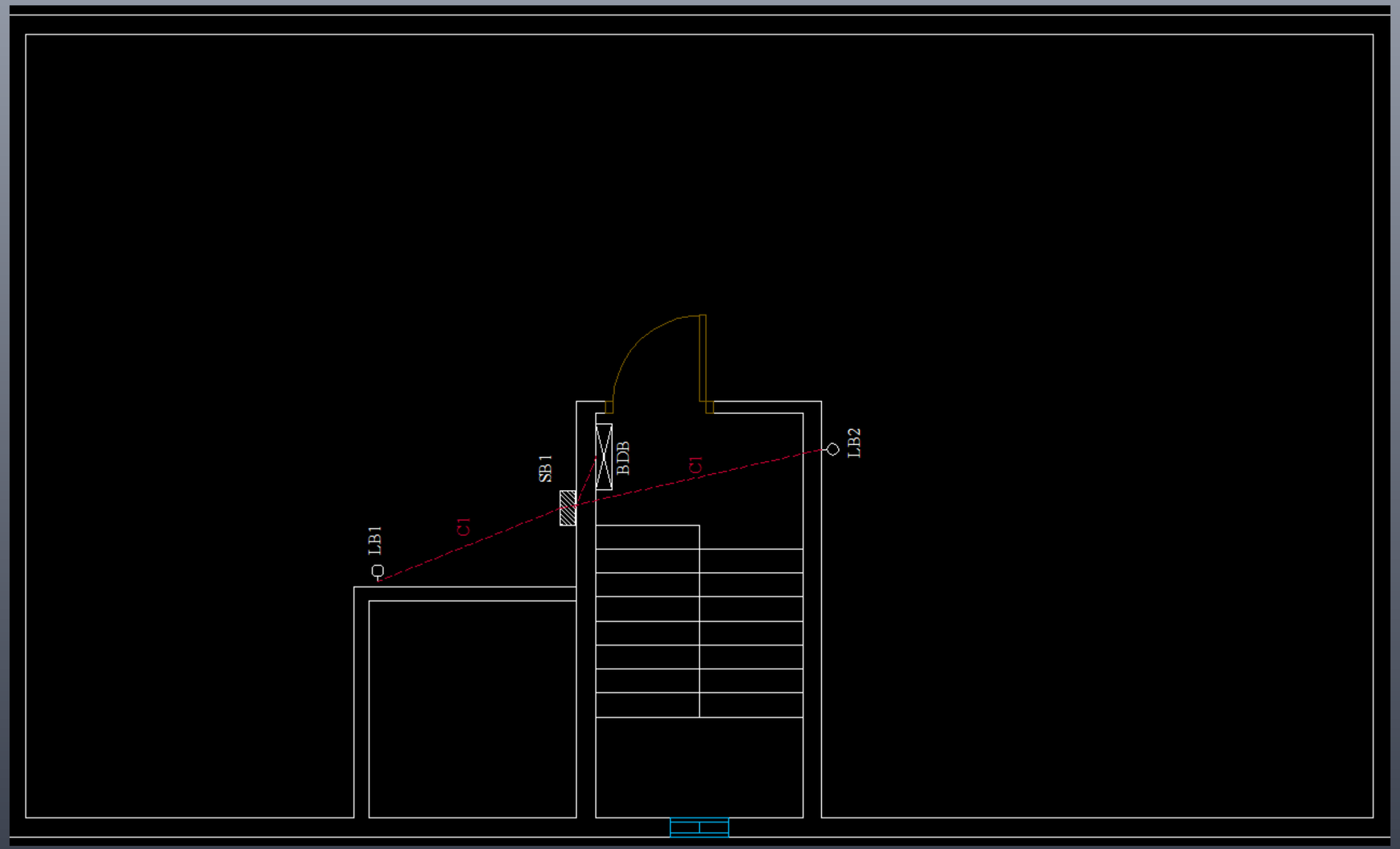
Ground floor — lighting and power circuits for the parking level

PARKING FLOOR WIRING

Switch boards	**SB1 – SB4**
Circuits	**CKT-1 to CKT-3**
Emergency board	**ESB1- 2**

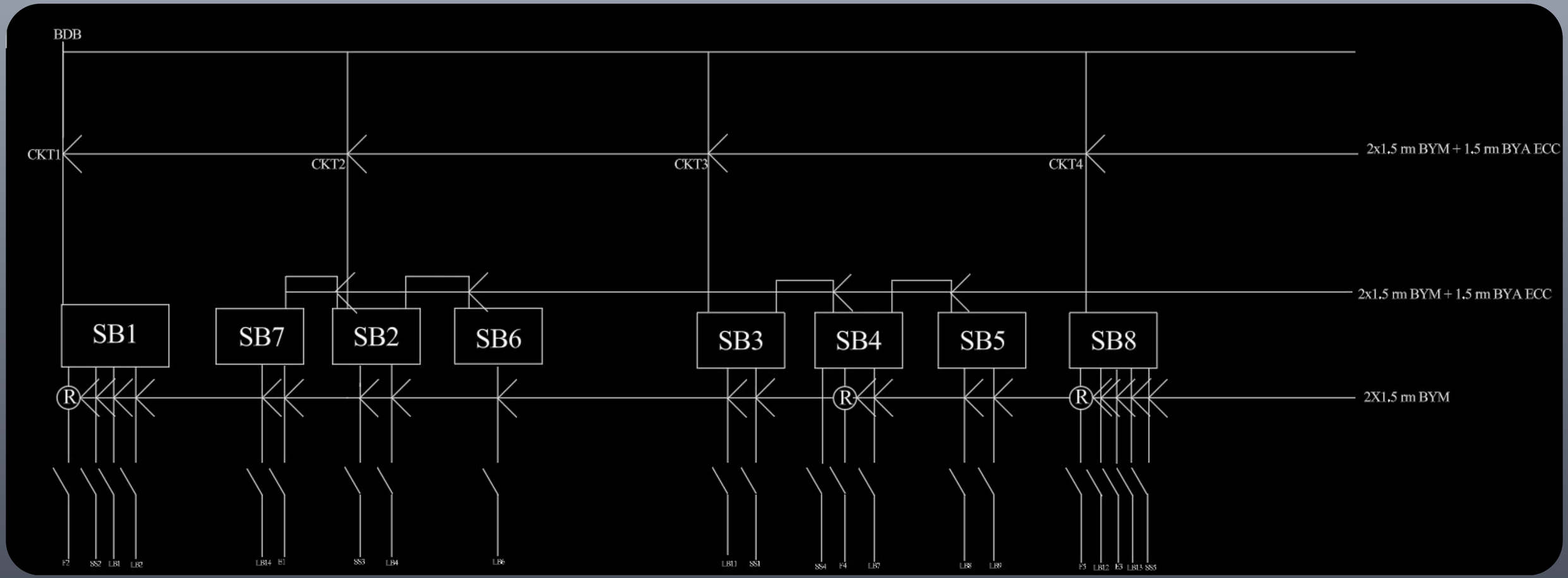
WIRING DESIGN

Wiring Layout — Roof Top



DISTRIBUTION SYSTEM

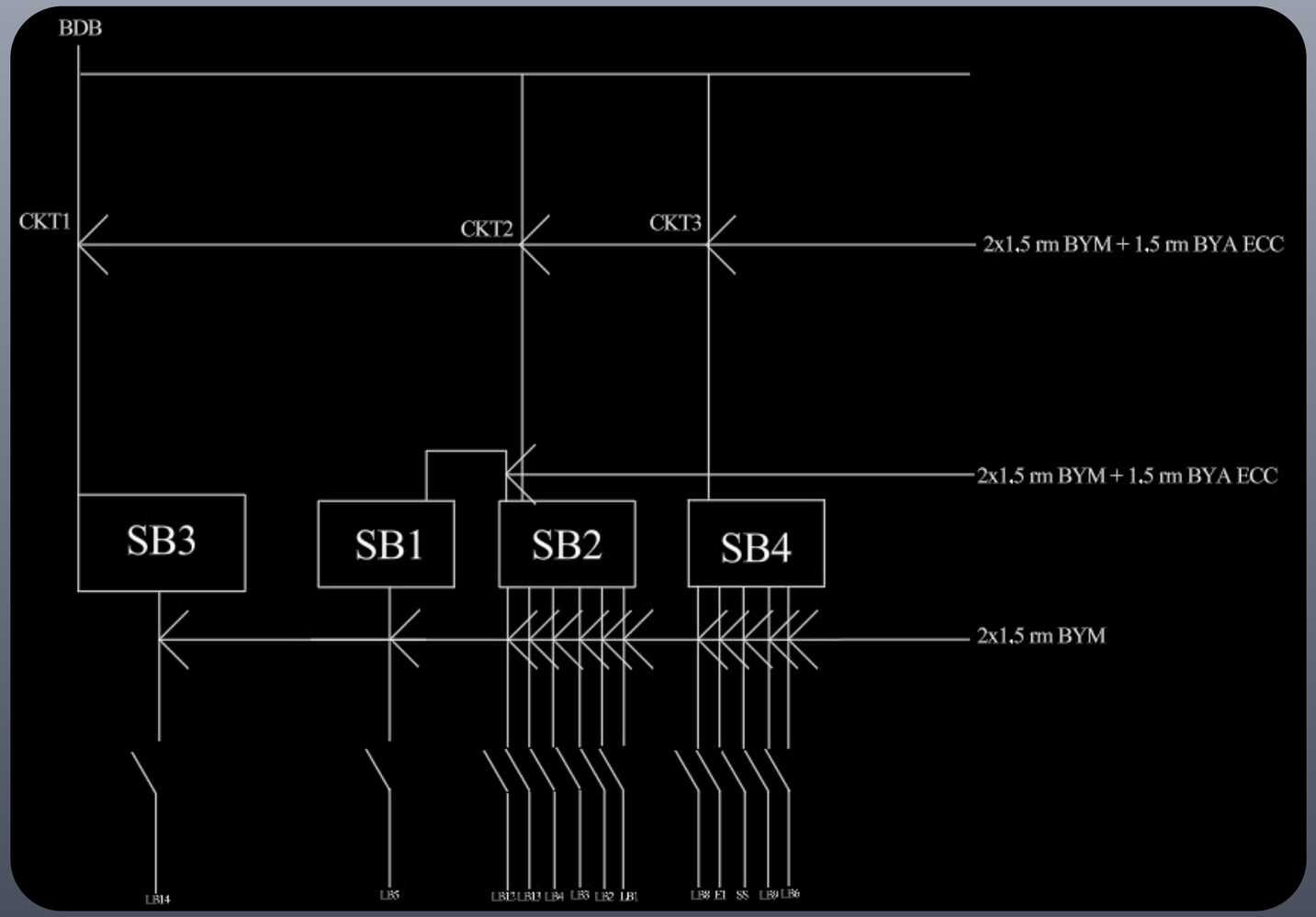
SB — Single Unit



04	08	04	1.5 mm²
CIRCUITS	SWITCH BOARD	CIRCUIT MCB	CONDUCTOR

DISTRIBUTION SYSTEM

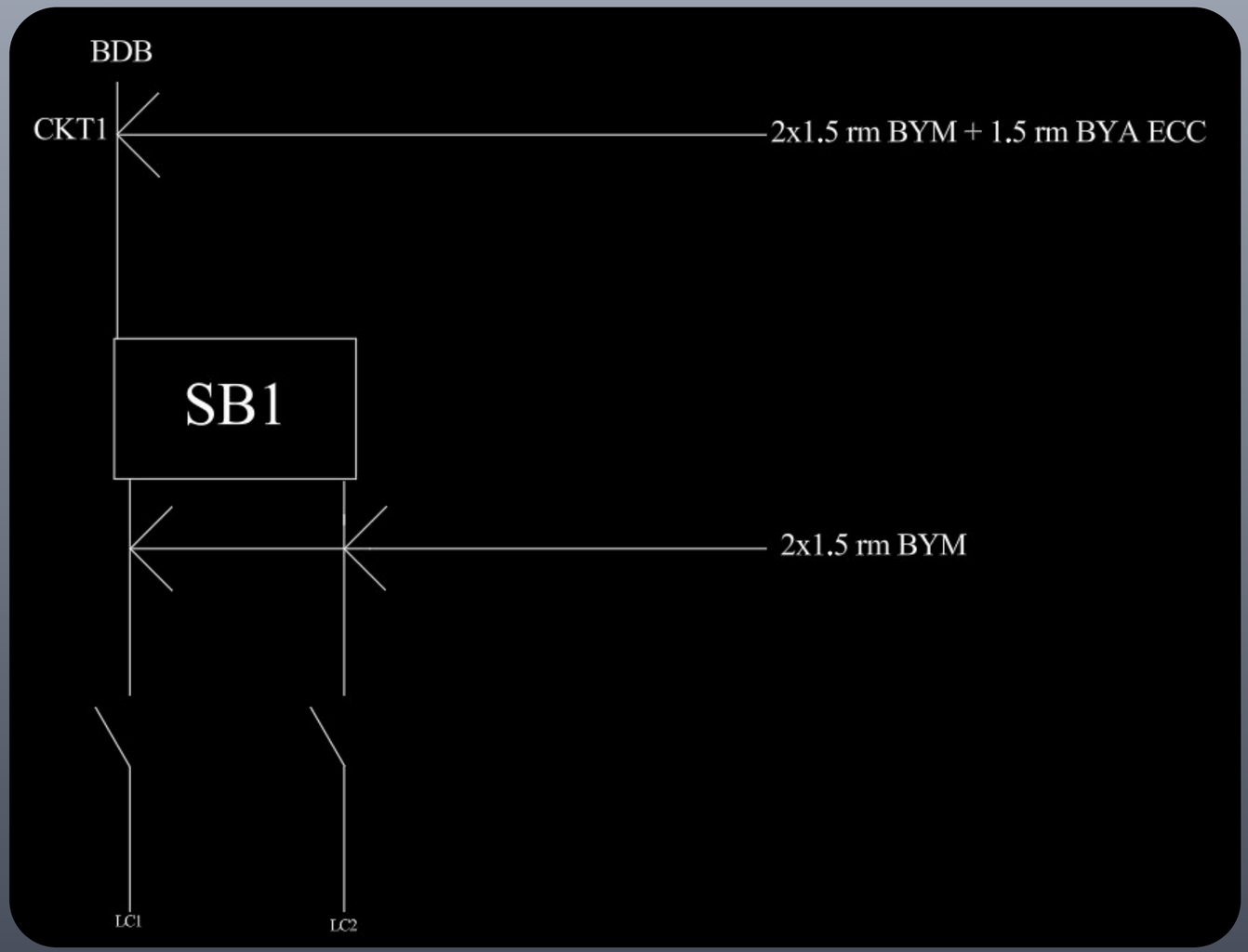
SB — GARAGE



03	04	04	1.5 mm²
CIRCUITS	SWITCH BOARD	CIRCUIT MCB	CONDUCTOR

DISTRIBUTION SYSTEM

SB — ROOFTOP



01
CIRCUITS

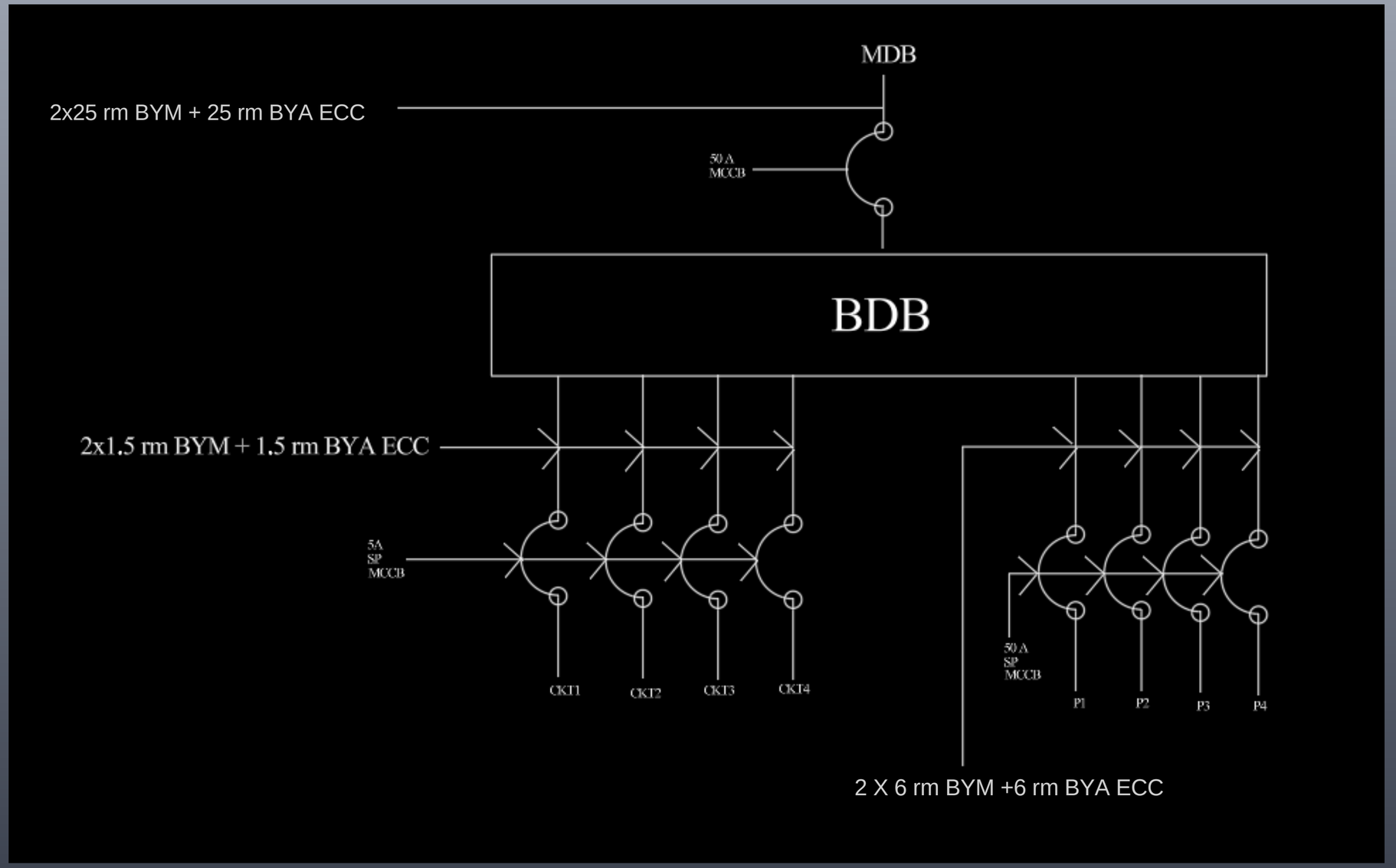
01
SWITCH BOARD

01
CIRCUIT MCB

1.5 mm²
CONDUCTOR

DISTRIBUTION SYSTEM

BDB — Single Unit



04

CIRCUITS

08

SWITCH BOARDS

5 A

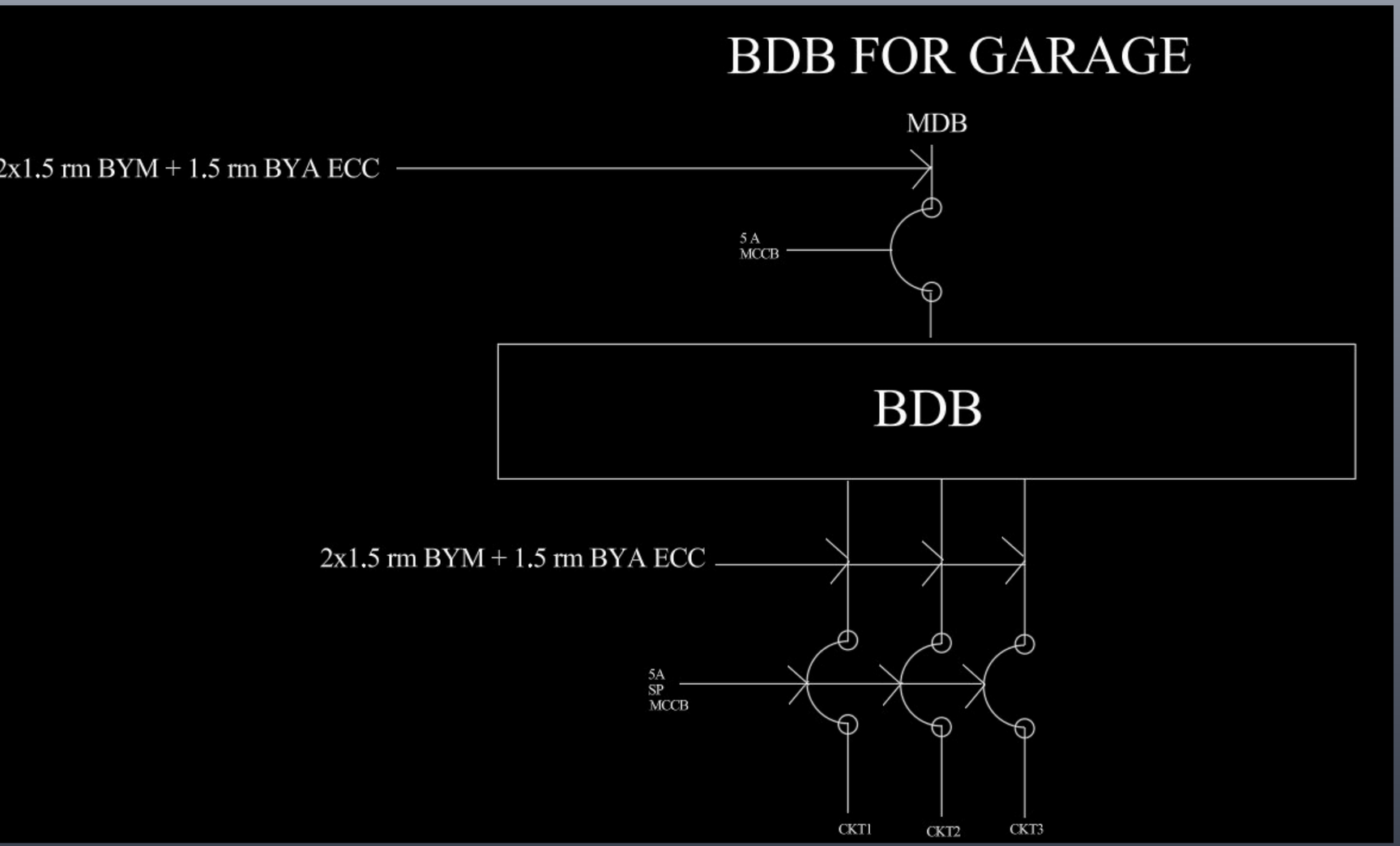
CIRCUIT MCB

1.5 mm²

CONDUCTOR

DISTRIBUTION SYSTEM

BDB — Parking Floor

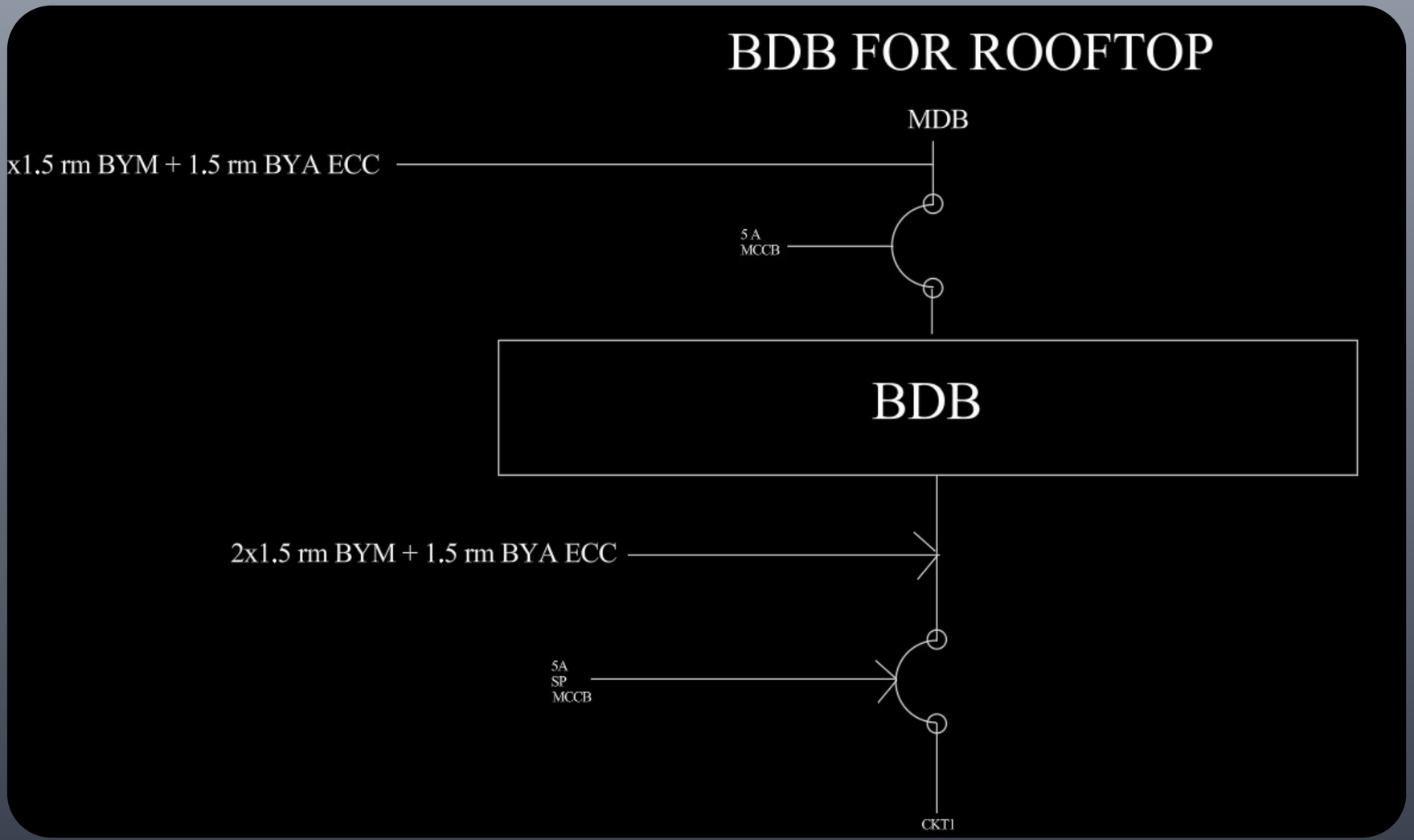


BOARD SUMMARY

Circuits	**3**
Sub-main	**2 × 1.5 mm² + ECC**
Final circuits	**2 × 1.5 mm² BYM**

DISTRIBUTION SYSTEM

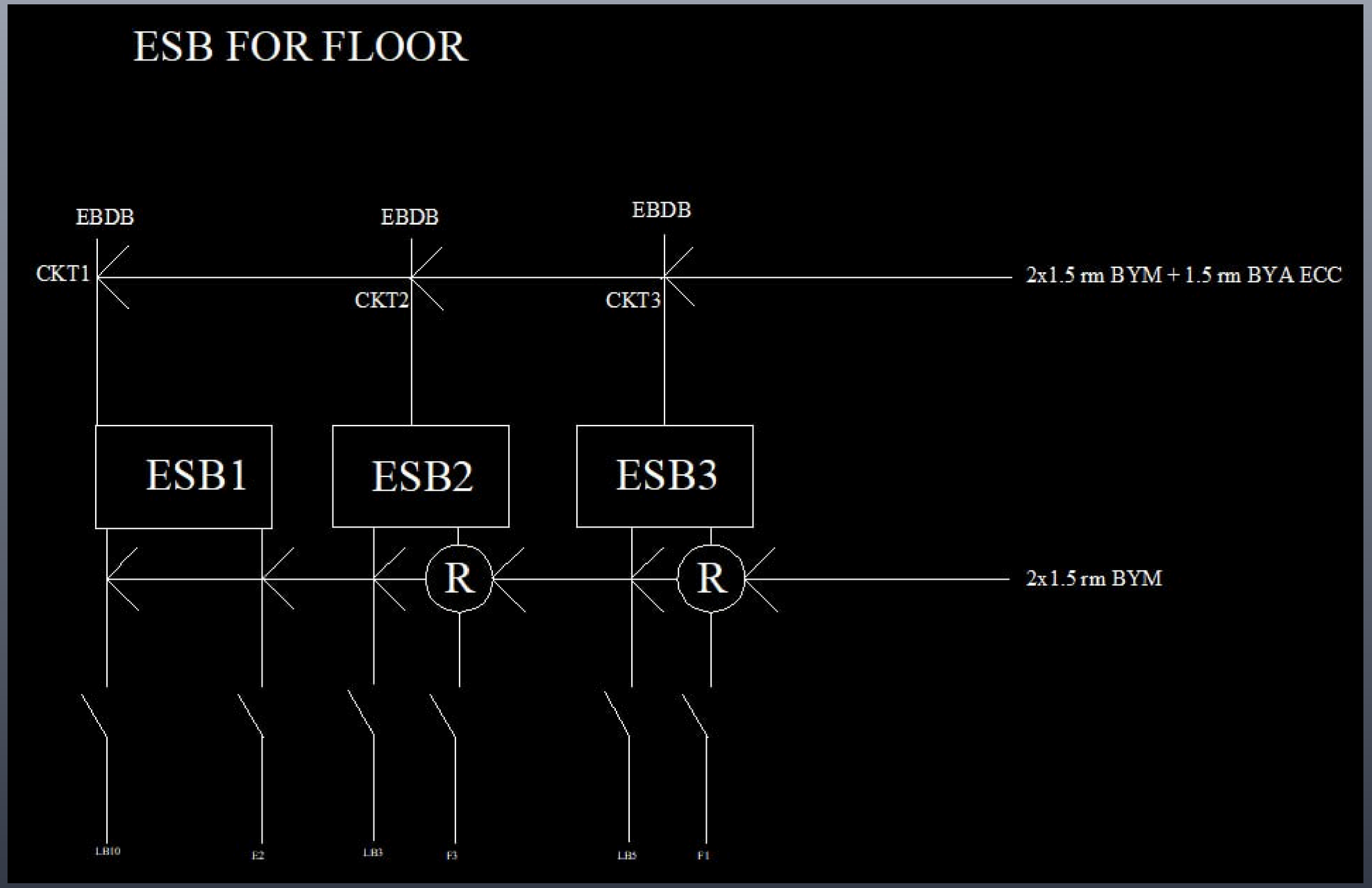
BDB — ROOFTOP



BOARD SUMMARY

Circuits	**1**
Sub-main	**2 × 1.5 mm² + ECC**
Final circuits	**2 × 1.5 mm² BYM**

ESB-SINGLE UNIT



03

CIRCUITS

03

SWITCH BOARDS

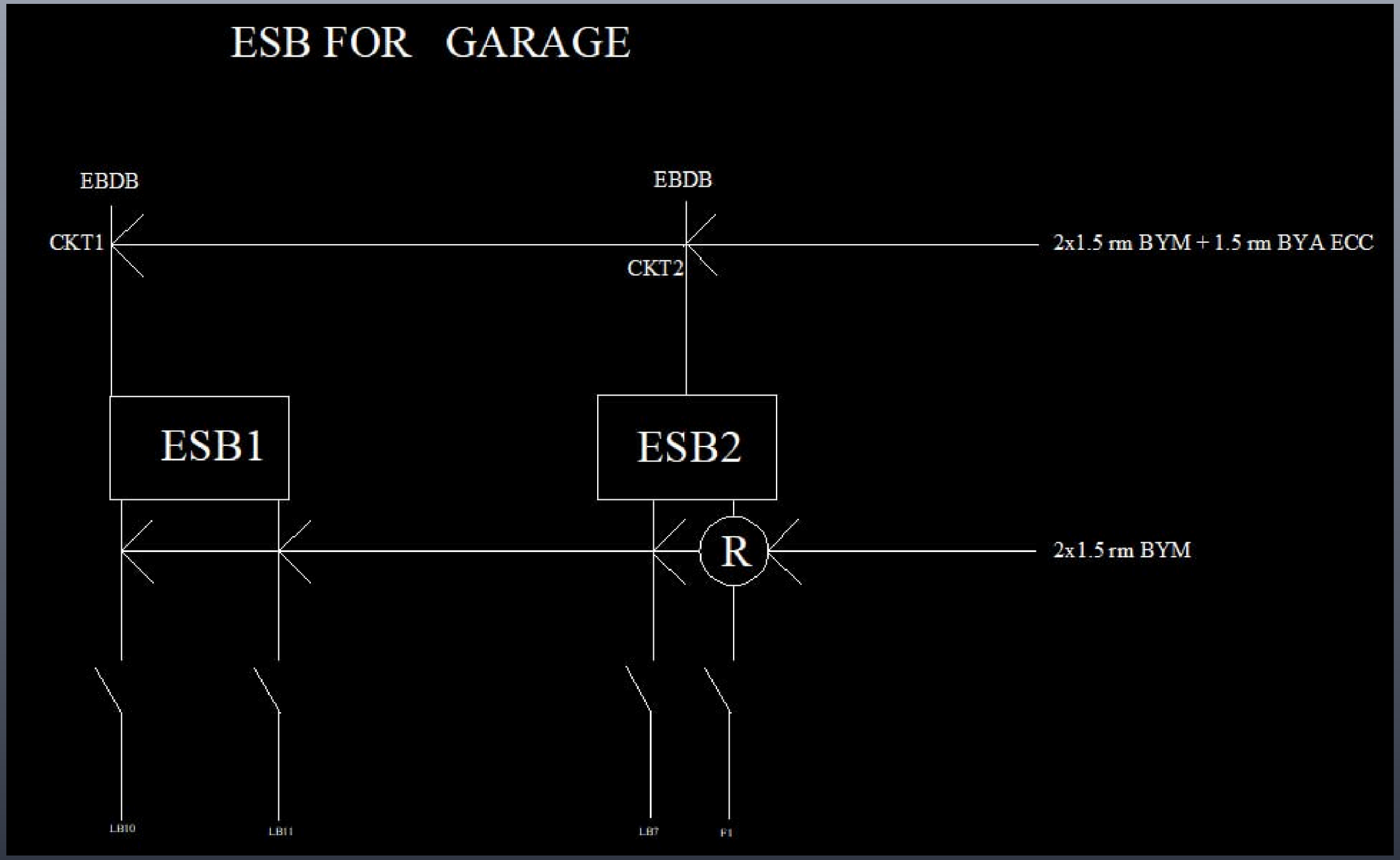
5 A

CIRCUIT MCB

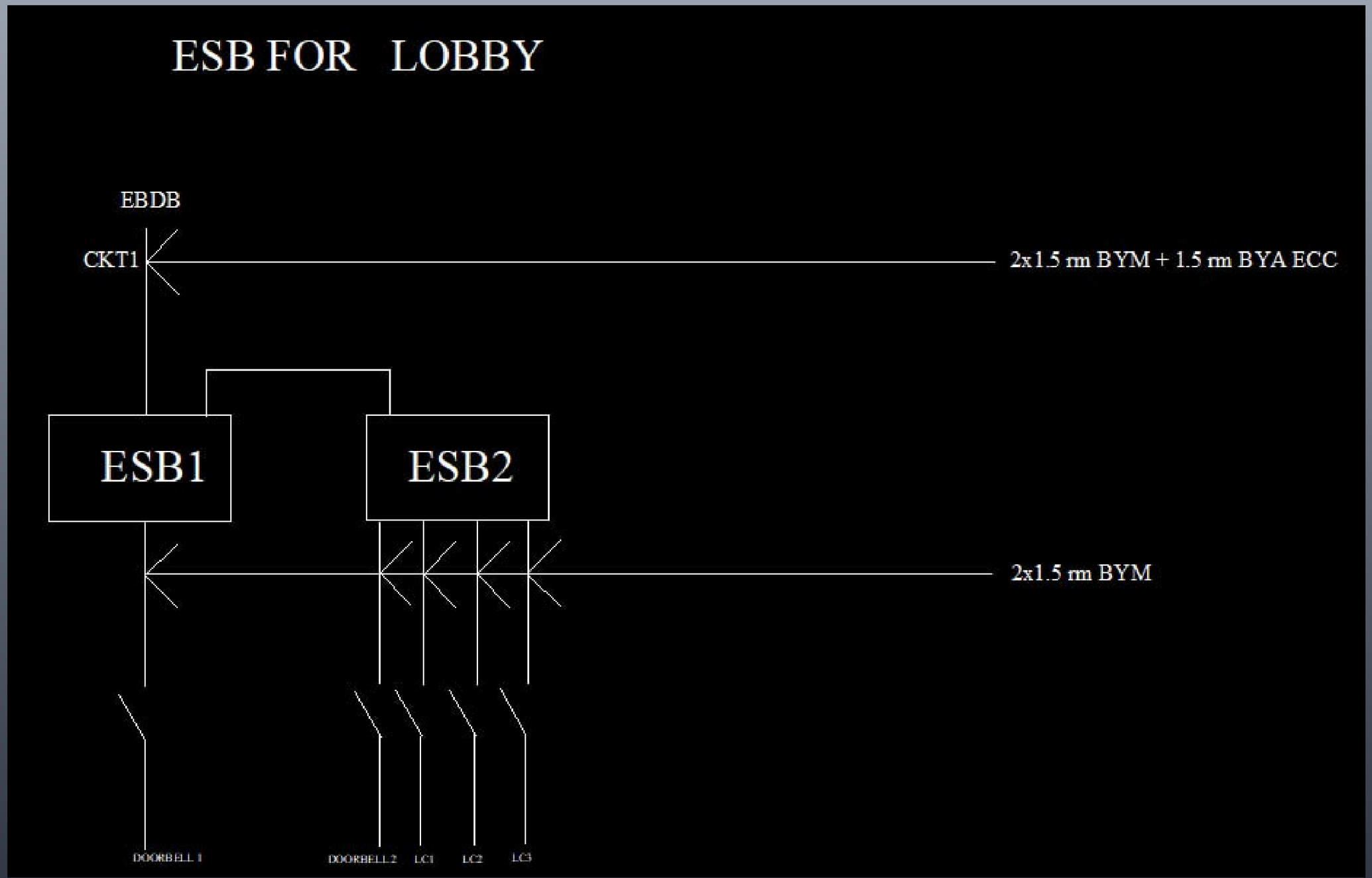
1.5 mm²

CONDUCTOR

ESB-GARAGE

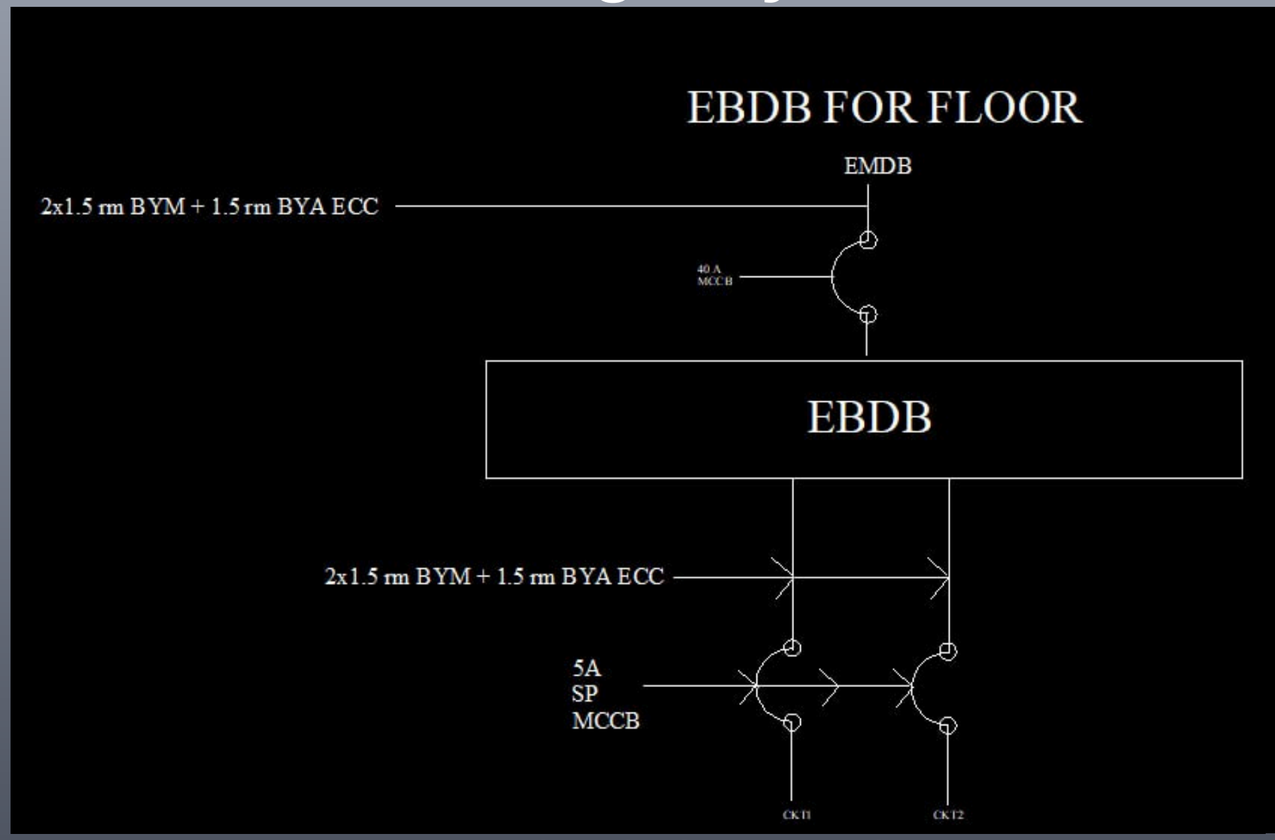

ESB FOR GARAGE
EBDB
EBDB
CKT1
CKT2
2x1.5 rm BYM + 1.5 rm BYA ECC
ESB1
ESB2
R
2x1.5 rm BYM
LB10
LB11
LB7
F1

ESB-LOBBY

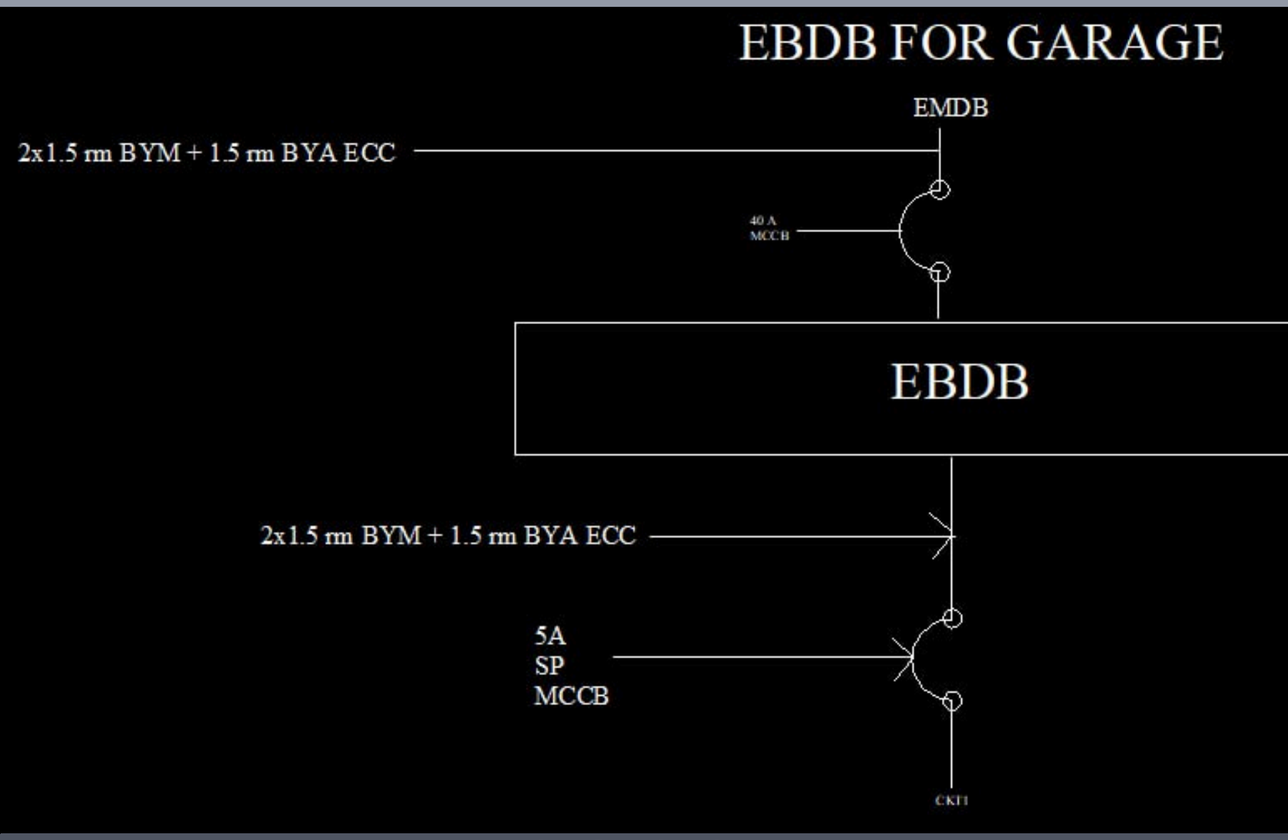

ESB FOR LOBBY
EBDB
CKT1
2x1.5 mm BYM + 1.5 mm BYA ECC
ESB1
ESB2
2x1.5 mm BYM
DOORBELL 1
DOORBELL 2
LC1
LC2
LC3

DISTRIBUTION SYSTEM

EBDB — Emergency Distribution Boards



ESDB — single unit : boards ESB 1, ESB 2 and ESB 3



ESDB — parking floor : board ESB 2

232.5 W

EMERGENCY LOAD

0.75

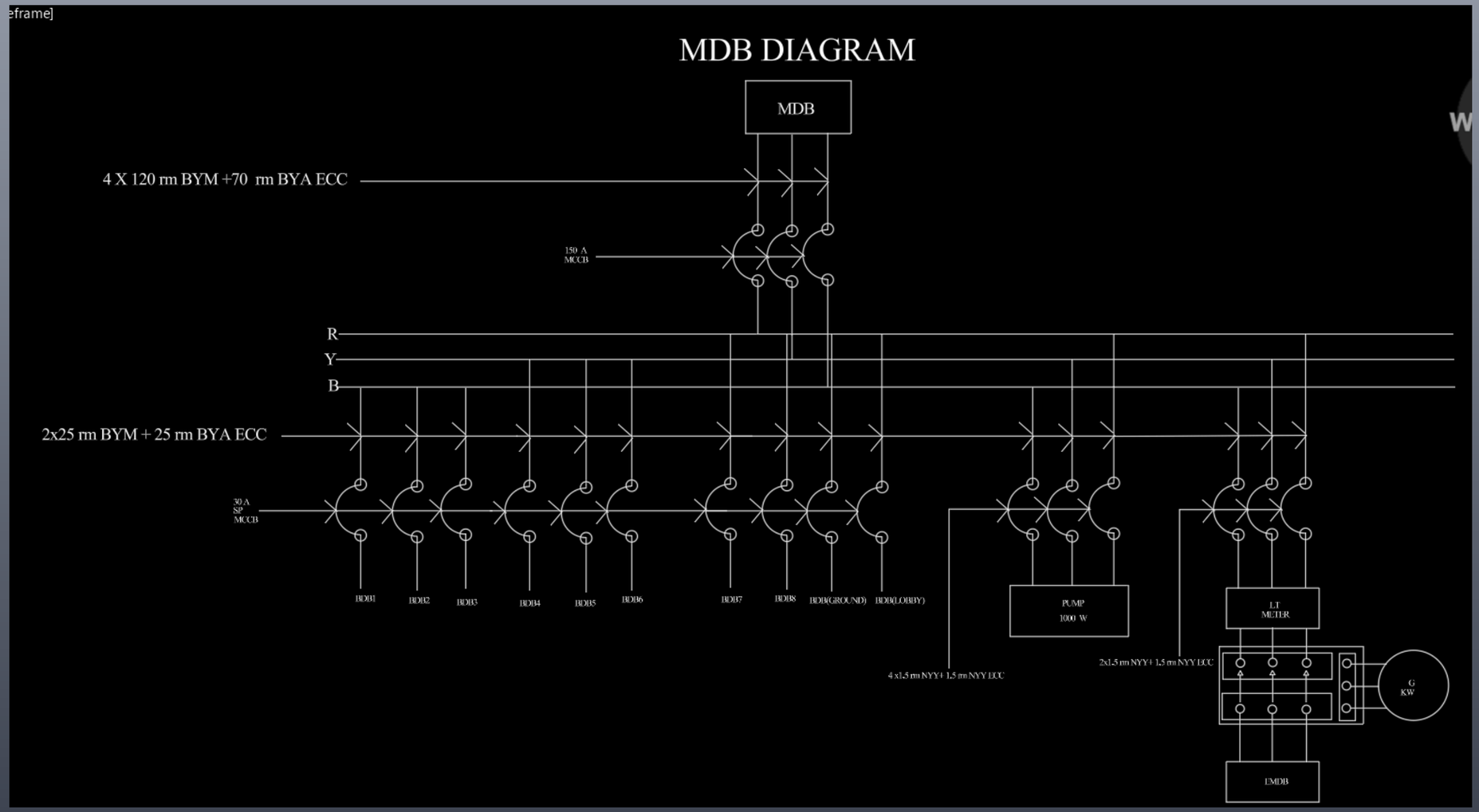
DEMAND FACTOR

1.5 A

ESDB CURRENT

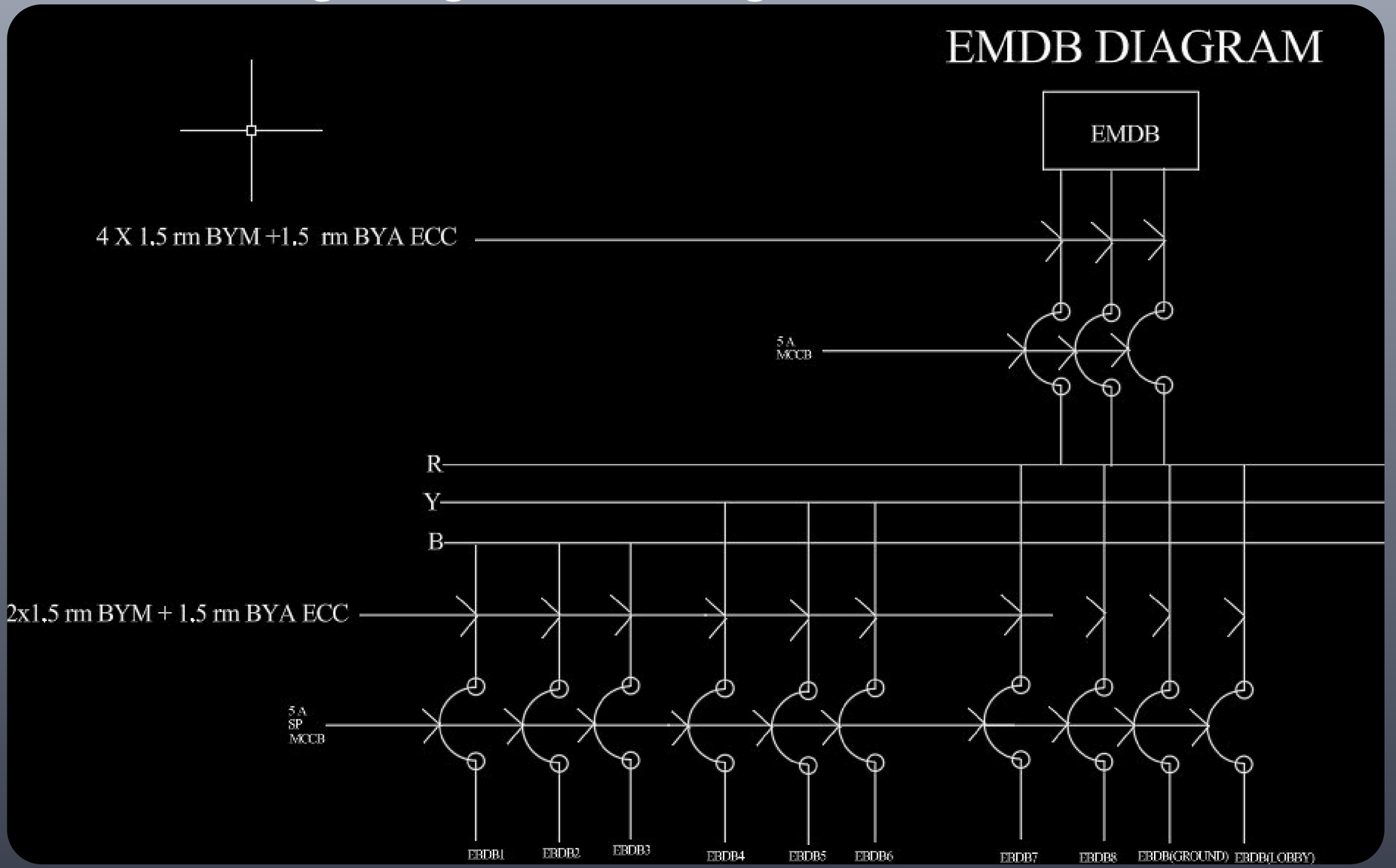
DISTRIBUTION SYSTEM

MDB — Building Single Line Diagram



Main distribution board — 3-phase incomer feeding all 8 branch boards, the pump, the EBDB and the standby generator

EMDB — Building Single Line Diagram



Emergency Main distribution board — 3-phase incomer feeding all 8 branch boards and the EBDB

LOAD SCHEDULE

Apparatus Schedule & Ratings

Apparatus	Rating	Per unit	Per floor	Building
Light point (LB / CB / TB)	20 W	14	28	112
Ceiling fan (F)	100 W	5	10	40
Switch board (SB)	—	8	16	80
Switchboard socket (SS)	100 W	5	10	40
3-pin socket outlet (P)	3000 W	4	8	32
Exhaust	50 W	3	6	24

ASSUMED RATINGS

Fan	**100 W**
Light bulb	**20 W**
Ceiling bulb	**20 W**
Switchboard socket	**100 W**
Exhaust	**50 W**

WHOLE BUILDING — 8 UNITS

112
LIGHT POINTS

40
CEILING FANS

72
SOCKET OUTLETS

24
Exhaust

LUMEN METHOD

$$N = (E \times A) \div (n \times F \times UF \times LLF)$$

used to fix the number of light points in each room from the required illuminance

BDB — Circuit Current of One Unit

$$I = P \div (V \cdot \cos \varphi) \quad \text{with } V = 220 \text{ V and } \cos \varphi = 0.7$$

Circuit	Connected apparatus	Load (W)	Current (A)
CKT-1	F + 2 × LB + SS	140	1.55
CKT-2	F + 3 × LB + E+ SS	210	1.36
CKT-3	4 × LB + F + 2 × SS	380	2.46
CKT-4	F + E +2 × LB + SS	290	1.88
	CKT total load	1020	43.92

Worked example — CKT-1 :

$$I = (100 + 2 \times 20) \div (220 \times 0.75) = 1.55 \text{ A}$$

Final conduit needed for each circuit connection wired in 2 × 1.5 mm^2 BYM + 1.5 mm^2 BYA ECC

Total BDB Load = Total Load from CKT × 0.75 + Total 3-Pin Socket Load × 0.5

Total Socket Load = 4 × 3000 = 12,000 W

Total BDB Load= 1020 × 0.75+12,000 × 0.5 = 765 + 6000 = 6765 W

I = BDB Load / (220 × 0.7)

I = 6765 / (220 × 0.7)

= 43.92 A

Therefore:

2 × 25 mm^2 BYM + 25 mm^2 BYA ECC used.

BREAKER & CABLE SELECTION

2.46 A maximum, so a 5 A SP MCB suits every lighting circuit

Socket circuits protected at 20 A for appliance loading

EBDB — Emergency Load & Current

ESDB load =(Total load × 0.75) + (Power socket load × 0.5)

Circuit	Connected apparatus	Load (W)
CKT-1′	E + LB	70
CKT-2′	F + LB	120
CKT-3′	F + LB	120
Total	**Emergency load of one unit**	31**0**

STEP BY STEP

1. Connected emergency load = 120 + 140 + 120 = 380 W
2. Apply the 0.75 demand factor :310 × 0.75 =232.5 W
3. I = 232.5 ÷ (220 × 0.75) = 232.5 ÷ 165 = 1.50 A
4. The needed conduit is 2 × 1.5 mm^2 BYM + 1.5 mm^2 ECC

RESULT — ONE UNIT

Connected load	**310 W**
Demand factor	**0.75**
Design load	**232.5**
ESDB current	**1.50 A**

WHOLE-BUILDING EMERGENCY

Per unit	**232.5 W**
× 8 units	**1860W**
Lobby	**80x4**

CALCULATION · 3 OF 3

MDB — Building Load & Breaker Sizing

I per phase = P per phase ÷ (V · cos φ) 3-phase supply, load split equally across R, Y and B — V = 220 V per phase, cos φ = 0.75

Segment of the building	Quantity	Load
Lobby	4	240W
Basement	1	120W
Floor	8	1860
Total EMDB		**2220W**

MDB Calculation:

P = BDB of floor × 8 + BDB of ground + Power of motor + EMDB + Power of roof

P = 6765 × 8 + 277.5 + 5000 + 2220 + 30

P = 61647.5 watt

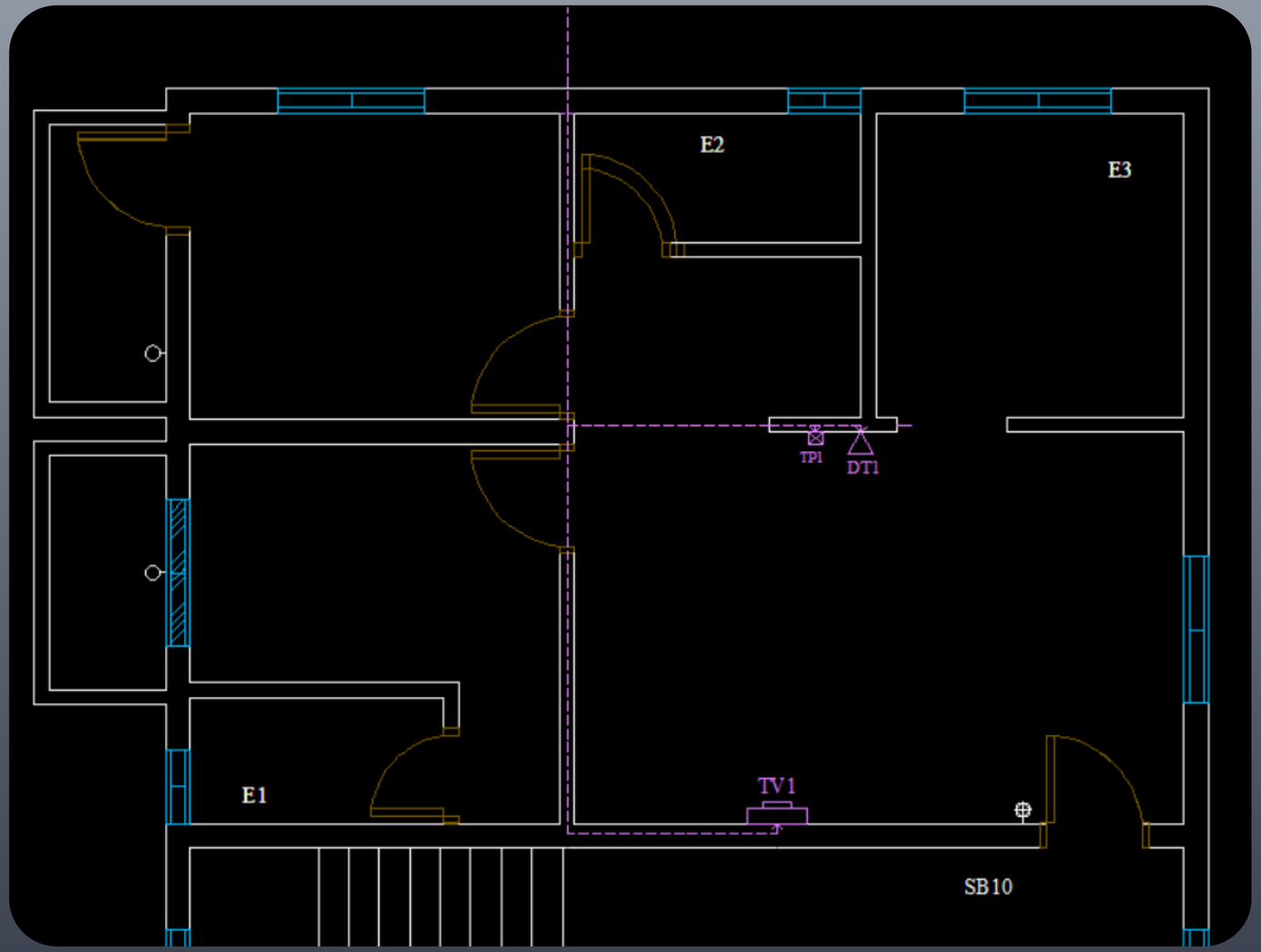
P = 3 V I pf → I = 133.43 A

conduit suitable for this is 4 × 120 mm^2 NYY + 70 mm^2 BYA ECC

RESULT

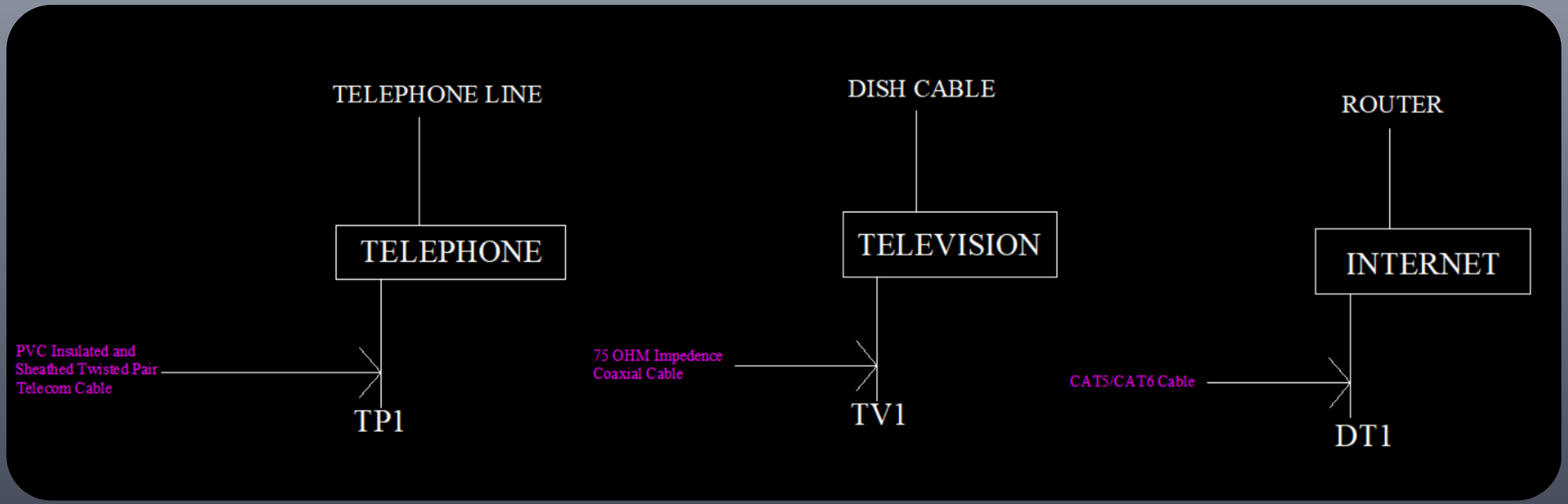
Total connected	**61.6 kW**
Phase current	**133.43 A**
Selected breaker	**150 A MCCB**

Conduit of TV, Telephone and Data Network (Single Unit):

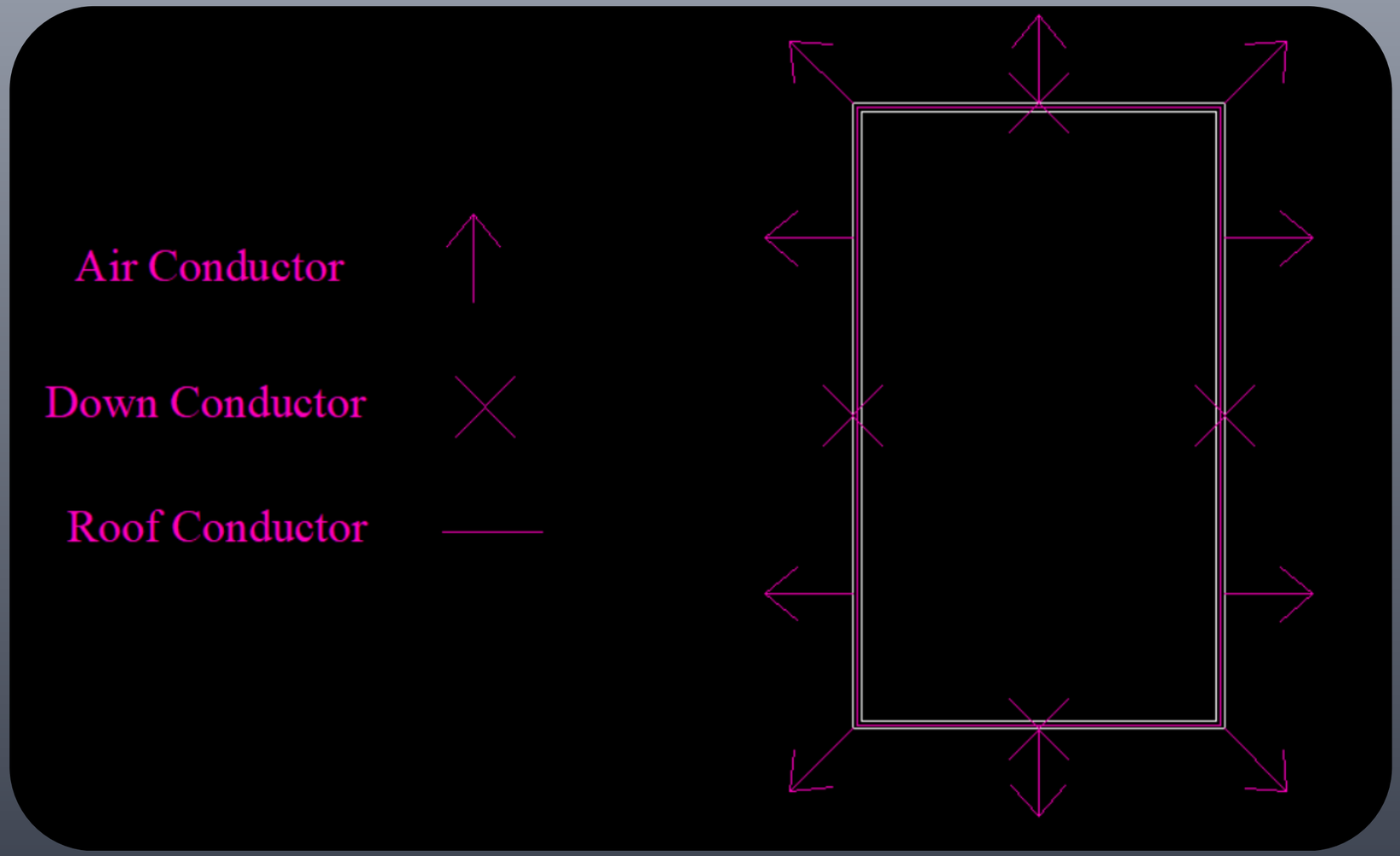


- TP1(Telephone Point) : PVC Insulated & Sheathed Twisted Pair Telecom Cable
- TV1 (Television Point) : 75-OHM Impedance Coaxial Cable
- DT1 (Data / Network Point) : CAT5 / CAT6 Cable

Distribution Board for Dish TV, Data Connection and Telephone Connection:



Lightning Protection System:



LPS CALCULATIONS:

- Building Perimeter=193.25'
- One air conductor is placed at each of the 4 corners of the roof.
- Maximum allowable distance between air conductors is maintained at 25'.
- Additional air conductors required to meet spacing rules: 1 along the width ceil(35.625/25}) - 1 = 1) and 2 along the length (ceil(61/25) - 1 = 2)
- Minimum down conductors required by perimeter is 2 (calculated as (193.25/100) = 1.9325).
- Down Conductors required 2, but for symmetry we used 4.

SELECTION OF CABLES & CIRCUIT BREAKERS SIZE

Table for Cables, Conduits, ECC, EL, Voltage drop and Current ratings of different specifications as per Manual of Eastern Cables, BICC cables and Tables, Electrical Conductors (International Standard Sizes) etc. :

A	B	C	D	E	F		G	H	I		J	
					a'	b'			a"	b"	a"'	B"'
3/0.029	1.5	5	16	10	6	10		27	27	22	16	20
7/0.029	2.5	10	16	10	4	7		16	36	30	22	28
7/0.036	4	15	14	10	3	5	1	10	47	39	30	37
7/0.044	6	20	14	10	2	4	1	6.8	59	50	38	47
7/0.052	10	30	10	10	1	2	1.5	4	78	68	52	63
7/0.064	16	40	10	10		1	1.5	2.6	100	94	70	85
19/0.052	25	50	6	6		1	2	1.6	130	125	91	110
19/0.064	35	60	6	6			2	1.2	155	160	112	136
19/0.072	50	70	6	6			2	0.93	185	195	136	164
19/0.083	70	100	1/0	1/0			2	0.65	225	245	173	207
37/0.072	95	120	1/0	1/0			2.5	0.48	270	300	216	253
37/0.083	120	150	1/0	1/0			2.5	0.4	310	350	244	291
37/0.093	150	200	1/0	1/0			3	0.34	350	405		333
37/0.130	185	250	3/0	3/0			3.5	0.29	390	460		381
61/0.093	240	300	3/0	3/0			4	0.24	450	555		452
61/0.103	300	425	3/0	3/0			4	0.22	515	640		526
91/0.093	400	585	3/0	3/0			6	0.2	586	770		639
91/0.103	500	685	3/0	3/0			6	0.18	680	900		752
127/0.103	630	800	3/0	3/0			6	0.17	800	1030		855

N OF CABLES & CIRCUIT BREAKERS SIZE (

A : Single core cable construction diameter, inch .... as per Imperial Standard Size : B.S.S (old).

B : Single core cable construction area, mm² .... as per Metric Standard Size : VDE.

C : CB designed current rating amps.

D : ECC (Earth Continuity Conductor), SWG.

E : EL (Earthing Lead), SWG

F : No. of cables in
- a') 3/4" diameter conduit
- b') 1" diameter conduit

G : GI pipe diameter (for 4 – core cable), inch.

H : Volt drop /amp/meter, Vd in mV (For PVC insulated, non-armoured single core cable 600/1000 volts as per BICC Metric Supplement, page 20-22, September 1969).

I : Maximum Current rating (For Type : NYY to VDE 0271/3, 69)
- a") 30° C ambient temperature, underground, amps
- b") 35° C ambient temperature in air, amps

J : Maximum current carrying capacity (For Type : BYA to B.S. 6004 : 1975)
- a"') Bunched & Enclosed in conduit, two cables single phase at 35° C, amps
- b"') Clipped to a surface or on a cable tray bunched and un-enclosed two cables single phase at 35° C, amps

NYY : PVC insulated and PVC sheathed cable, rated voltage 600/1000 volts.

BYA : PVC insulated non-sheathed single core cable, rated voltage 450/750 volts.

Thank you for your attention